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Practice Assignment • 10 min

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Module 4 Introduction

Lesson 1: Piecewise polynomials

- **Reading:** Piecewise Polynomials Readings
  1h
- **Video:** Piecewise Polynomials, the Method and Theory
  5 min
- **Practice Assignment:** Piecewise polynomials Quiz
  Submitted
- **Lab:** Coding Example
  1h
- **Lab:** Coding Exercise
  1h

Lesson 2: Smoothing Splines

Lesson 3: Regularization via Reproducing Kernel Hilbert Spaces

Module 4 Summative Assessment

Module 4 Summary

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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Instructions

1. What is a key characteristic of piecewise polynomial methods in modeling? **1 / 1 point**

○ They only use linear functions for modeling.
● They involve using different polynomial equations over different ranges of the independent variable.
○ They rely on a single polynomial equation for the entire range of data.
○ They do not require continuity between segments.

Correct
Correct! Piecewise polynomial methods use separate polynomials in different segments of the data.

2. What is the primary advantage of using piecewise polynomial methods compared to global polynomial fitting? **1 / 1 point**

○ They eliminate the need for data preprocessing.
○ They always result in simpler models with fewer parameters.
● They can provide a better fit for data with varying trends or behaviors in different ranges.
○ They are always computationally more efficient.

Correct
Correct! Piecewise polynomials adapt to changes in data patterns across different segments.

3. What is one challenge associated with piecewise polynomial regression? **1 / 1 point**

○ It requires a different dataset for each polynomial piece.
○ Piecewise polynomial regression cannot handle categorical variables.
● Choosing the number and location of segments can be complex and may require domain knowledge or validation methods.
○ It is not suitable for datasets with more than 100 observations.

Correct
Correct! Determining the appropriate knots is a crucial and challenging aspect of piecewise polynomial modeling.

Piecewise polynomials Quiz

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice **Let's chat**

Assignment details

Submitted	**Attempts**
Nov 16, 1:05 AM IST	Unlimited

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Your grade
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View submission **See feedback**

100%

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